

A Load-Bearing Standard for Representation Claims

Concern-weighted transport, gauge fixing, and commitment effect as evidence obligations for interpretability and safety

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ABSTRACT

Interpretability, causal representation learning, and AI safety keep making the same class of claim: *the system has a representation of X* — a feature, a concept, a goal, a self-model, a remembered instruction. Most such claims are supported by evidence of **availability** (X is decodable, a probe recovers it, an activation correlates with it) rather than evidence of **use** (X causally controls what the system commits to). These are not the same, and the gap is where deployment surprises, unfaithful explanations, and mislocated mechanisms live.

This paper proposes a minimal standard that a representation claim must meet to count as *load-bearing*. The standard has four obligations: a **concern measure** stating which distinctions matter and how much; a **transport chain** stating which contexts, layers, or timesteps the distinction must survive; a **gauge-fixing intervention** that separates the claimed description from observationally equivalent alternatives; and a **commitment effect** showing that changing the distinction changes what the system does. These combine into a single bookkeeping inequality, $\text{Load}_\gamma(h) \geq (\alpha - \delta)\epsilon$, read not as a deep theorem but as an accounting identity that refuses to be satisfied unless all four fields of evidence are supplied. We state the standard, derive two operational laws (the *moved-bottleneck law* and the *decodability-is-not-load law*), give a worked synthetic demonstration and results for each, and provide a seven-step protocol plus an application to chain-of-thought faithfulness in long-horizon agents. Scope is deliberately narrow: this is a measurement discipline for representation claims, not a theory of intelligence, and its empirical support here is synthetic and illustrative.

1. The problem

A representation claim is usually stated as a question about *location*: where is the feature, the concept, the goal, the self-model? This phrasing invites the wrong evidence. A direction in activation space can be linearly decodable at high AUC while contributing nothing to the output. A

neural signal can track a variable it does not control. A stated chain of reasoning can name a fact the model did not use to answer. In each case the representation *exists as an object* and yet is not *doing the work* the claim implies.

The failure recurs across fields in different costume:

- **Machine learning.** Two models with identical validation scores diverge under deployment shift; the held-out metric did not identify which distinction each model actually relies on (*underspecification*, D'Amour et al. 2022).
- **Causal representation learning.** Observational data leaves a coordinate gauge: a latent axis can be predictive without being the causal factor, and unsupervised disentanglement is impossible without inductive bias (Locatello et al. 2019; Schölkopf et al. 2021).
- **Mechanistic interpretability.** A feature can be decoded (weak existence test) without participating in the circuit that produces the answer (Olah et al. 2020; Geiger et al. 2021).
- **Safety.** An agent can emit the correct rationale while the operative bottleneck has moved to memory, a tool, or a later action surface.

The common fix is to stop asking *where* and start asking *when*: **when does a distinction survive the transformations that matter, and when does changing it change the commitment?** The rest of this paper makes that question into a checkable standard.

2. The standard

We work in a finite setting for cleanliness; the continuous version replaces sums with integrals and maps with Markov kernels without changing the logic.

Contexts and sections. Let $\mathcal{C}$ be a finite set of *contexts* — a context is any place a distinction can live: a data distribution, a task environment, a model layer, a time slice, a sensorimotor condition, an intervention regime, or an action surface. For each context c , let $\mathcal{S}(c)$ be a finite set of *local sections* (possible assignments of the distinction there). A distinction h at c has an **extension** $Z_h(c) \subseteq \mathcal{S}(c)$: the sections compatible with h .

Concern. A *concern density* $\kappa_c : \mathcal{S}(c) \rightarrow \mathbb{R}_{\geq 0}$ assigns nonnegative weight to sections according to how much they matter for the system's future control, viability, action, or evaluation. "Concern" is not a folk-psychological attitude; it is a measure. The **concern-weighted mass** of h is

$$W_c(h) = \sum_{s \in Z_h(c)} \kappa_c(s).$$

When $\kappa_c \equiv 1$, $W_c(h) = |Z_h(c)|$ – plain extension count. This is the uniform special case, and it is the case that ordinary average-validation selection silently assumes.

Transport. Each morphism $m : c \rightarrow d$ carries a *transport* operation T_m (a map $S(c) \rightarrow S(d)$, or a Markov kernel for stochastic systems). Along a chain $\gamma = (c_0 \rightarrow \cdots \rightarrow c_n)$, mass can only be lost at each monotone step; the *transport loss* is $\ell_i = W_i - W_{i+1} \geq 0$, and telescoping gives

$$W_n(h) = W_0(h) - \sum_i \ell_i(h) \geq W_0(h) - \delta \quad \text{if } \sum_i \ell_i \leq \delta.$$

The point of transport is temporal: a distinction with high concern mass at perception can have *zero* mass at the action surface if it is filtered out before commitment.

Gauge. Let a *gauge group* G_c act on a latent description space $L(c)$ with observation map $O_c : L(c) \rightarrow S(c)$. A property is **gauge-ambiguous** when observations (and commitments) are invariant under a nontrivial $g \in G_c$ that nonetheless changes the property. Formally: if $O_c(l) = O_c(g \cdot l)$ and the commitment is invariant, then no procedure reading only observations and commitments can tell l from $g \cdot l$. Decodability does not break the gauge; an **intervention** that separates the induced evidence by at least η does. This is the formal content of "correlation is not mechanism": absent an intervention that distinguishes two descriptions, their difference is not yet identified.

Commitment. A *commitment surface* is the context c_n where future control is fixed, equipped with a commitment map $a_n : S(c_n) \rightarrow A$ (actions, answers, policies, reports, motor commands, decisions) and a value $V : A \rightarrow \mathbb{R}$. The **commitment effect** of h is

$$\text{CE}(h) = | \mathbb{E}[V \mid \text{do}(h)] - \mathbb{E}[V \mid \text{do}(-h)] |,$$

the change in outcome under enforcing versus contrasting the transported distinction.

Load. Define $\text{Load}_\gamma(h) = W_n(T_\gamma h) \cdot \text{CE}(T_\gamma h)$. If (1) initial concern mass $W_0 \geq \alpha$, (2) transport loss $\sum_i \ell_i \leq \delta < \alpha$, (3) an intervention family gauge-fixes the property at scale η , and (4) the gauge-corrected commitment effect $\text{CE} - \eta \geq \varepsilon > 0$, then

$$\text{Load}_\gamma(h) \geq (\alpha - \delta) \varepsilon$$

and h is ε -load-bearing at concern scale α relative to γ , the intervention family, κ , and the commitment surface.

On the status of the inequality. It is a product of two nonnegative lower bounds; the proof is one line and carries no predictive content. This is intentional. The value is not the mathematics but the *bookkeeping it forces*: the inequality cannot even be written down unless the claimant has named a concern measure, measured transport loss, exhibited a gauge-fixing intervention, and measured a commitment effect. It converts "the system has X " from a rhetorical assertion into a claim with four inspectable fields. Read it as a definition of terms and an evidence contract, not a theorem.

3. Two operational laws

The standard yields two rules that are directly usable at the bench.

Law 1 — Decodability is not load-bearing. A distinction can satisfy availability (X is decodable, high probe AUC) while $CE(X) \approx 0$. Availability is necessary but not sufficient; only a gauge-fixing intervention with a nonzero commitment effect establishes use. Probing is a weak existence test; patching, ablation, interchange intervention, and model editing are the gauge-fixing and commitment tests.

Law 2 — The operative representation is at the moved bottleneck. In systems with memory, tools, recurrence, or delegation, the distinction that controls the commitment is often not where it first appears or where it is most legible. The load-bearing location is the bottleneck whose intervention changes the commitment — which may have moved downstream to a scratchpad, a tool call, a retrieved context, or a policy prior. Locate the mechanism by *where patching changes the output*, not by where the information is easiest to read.

Both laws are falsifiable. The framework is wrong anywhere raw decodability reliably identifies mechanism across interventions, or wherever the earliest / most-legible encoding is reliably the operative one.

4. Worked demonstrations

Two synthetic constructions with known ground truth illustrate the laws. These are *plausibility checks* — the data is built to have the property under test, so they show the obligations can be operationalized and measured, not that the framework predicts anything not built in.

Demo A — Decodability $\neq$ load (Law 1). Construct a task with a causal hidden feature and a highly decodable distractor; both predict the label, but only the causal feature enters the commitment logit. Measure two quantities separately: probe AUC (availability) and average patch

effect (commitment). In the reference run the distractor reaches probe AUC ≈ 0.999 while its patch effect is ≈ 0.010 ; the causal feature's patch effect is $\sim 36\times$ larger. The two axes dissociate exactly as Law 1 predicts. The methodological lesson: *report both axes and never let a probe stand in for a patch.*

Demo B — Moved bottleneck (Law 2). Simulate a long-horizon system whose final answer depends on either scratchpad state or tool state. Early memory can be present and interpretable while no longer controlling the output. Patch each candidate bottleneck separately and compare commitment change. In the reference run, patching the active bottleneck changed final behavior while patching the early encoding or inactive storage did not (active/inactive effect ratio $\gg 1$; bottleneck localization ≈ 0.93). The lesson: *sweep candidate locations with interventions and let the commitment effect localize the mechanism.*

The same construction pattern extends to concern-weighted OOD model selection, causal gauge fixing via paired interventions, and refference / self–world attribution via efference-copy nulls — each a controlled test of one obligation of the standard.

4.1 Results

The full suite ran as a five-track synthetic battery on NVIDIA L4 workers (max_containers=64, 64 seeds per track, 320 remote cells; Modal app ap-GFeFvaDvcvaKG0jWyeciX8). Each track targets one obligation of the standard and has an explicit built-in failure mode; a track *passes* when the obligation-satisfying estimator beats the obligation-blind baseline on the pre-registered metric. All five passed. Figures below are as reported for that run.

Track	Obligation tested	Key metric	Result	Gate
Concern-weighted OOD	Concern weighting changes hypothesis selection	weighted deployment-error gain; stable-hypothesis selection rate	0.458; chosen in 1.000 of seeds	PASS
Causal gauge fixing	Intervention identifies latent gauge	factor-alignment lift; commitment-effect lift	0.494; 0.164	PASS
Mechanistic commitment	Decodability $\neq$ use	distractor probe AUC vs. patch effect; causal/distractor patch ratio	AUC 0.999, patch 0.010; 35.96 $\times$	PASS
Reafference / null	Self-world attribution needs an efference-style null	attribution-AUC lift; correction-error reduction; null-intervention AUC	0.331; 0.669; 0.964	PASS
Moved bottleneck	Operative representation is where patching moves the output	active-vs-early patch gain; active/inactive effect ratio; localization	0.488; 69.42 $\times$; 0.930	PASS

Two numbers carry the paper's core claims. In the mechanistic-commitment track a distractor is nearly perfectly decodable (probe AUC 0.999) yet almost inert at the output (patch effect 0.010), while the causal feature's patch effect is $\sim 36\times$ larger — a clean dissociation of availability from use (Law 1). In the moved-bottleneck track, intervening on the active bottleneck moves the commitment $\sim 69\times$ more than intervening on early or inactive storage, and localization reaches 0.930 (Law 2).

A conservative timeout-based budget guard estimated \$63.94 of a \$250.00 ceiling before dispatch. This is a cost note, not a scientific result, but it establishes that the whole battery is cheap enough to rerun on demand. **Provenance:** values above are as reported in the source manuscript's run; they have not been re-derived from the raw results JSON in this document.

5. Protocol

To apply the standard to a new representation claim:

1. **Name the commitment surface.** Which action, answer, policy, or decision is at stake?
2. **Define the concern density.** Which distinctions matter for that commitment, with what weights? (Deployment consequence, not training salience.)

3. **Specify the transport chain.** Through which contexts, layers, times, or interfaces must the distinction pass to reach the surface?
4. **List the gauge alternatives.** Which different latent descriptions produce the same observations?
5. **Choose gauge-fixing interventions.** What perturbation, ablation, environment shift, patch, or matched contrast separates the alternatives?
6. **Measure the commitment effect.** Does changing the distinction change the commitment, weighted by concern?
7. **Report the bound.** State W_0 , transport loss, gauge-fixing scale, commitment effect, and the resulting **Load** lower bound.

A claim that cannot fill steps 4–6 is an availability claim, not a load-bearing one, and should be labeled as such.

6. Application: chain-of-thought faithfulness in long-horizon agents

The standard maps cleanly onto a live safety problem: does an agent's stated reasoning actually control its action, or is it a post-hoc narration? In these terms, the claim "the model used reasoning step h " is a load-bearing claim, and the four obligations become a faithfulness eval:

- **Commitment surface:** the final action or answer.
- **Concern:** weight reasoning steps by their consequence for that action, not by their prominence in the transcript.
- **Transport chain:** instruction $\rightarrow$ intermediate reasoning / scratchpad $\rightarrow$ tool state / retrieval $\rightarrow$ final decision.
- **Gauge alternatives:** the operative information may sit in the weights, the visible context, a tool, or a policy prior — all consistent with the same transcript.
- **Gauge-fixing intervention:** move, corrupt, or patch each candidate step or store separately (edit the stated step; ablate the scratchpad token; swap the tool result) with matched controls.
- **Commitment effect:** measure whether the final behavior changes.

Law 2 predicts the sharp case: a model can state a reason whose intervention leaves the output unchanged (unfaithful) while an unstated bottleneck downstream carries the real load. A faithfulness result therefore requires a commitment-effect measurement at the moved bottleneck,

not a plausibility read of the transcript. This is a concrete, fundable eval design rather than a general appeal to "interpretable reasoning."

7. Scope and non-claims

The contribution is a measurement discipline, and its limits should travel with it. The inequality is elementary bookkeeping, not a predictive theorem; its worth is in what it forces one to measure. The demonstrations are synthetic and constructed to exhibit the property under test — they validate operationalizability, not human, neural, biological, or foundation-model generality. Concern is treated here as a scalar for entry-level clarity; many systems need vector-valued or context-dependent concern. Nothing here implies that a load-bearing distinction is conscious, semantic, or morally relevant — those require separate arguments. And the standard does not replace experiment: it states what must be measured; the interventions decide the case.

What the standard does buy is a single, checkable answer to a question three fields keep answering by assertion: *when has a distinction earned the right to be called a representation the system uses?* When its concern-weighted mass survives transport to the commitment surface, its gauge is fixed by an intervention, and changing it changes what the system does.

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